MIL-B-20467A(PA)
14 February 1974
SUPERSEDING
MIL-B-20467
27 November 1951

MILITARY SPECIFICATION

BOXBOARD
(FOR USE IN AMMUNITION)

* 1. SCOPE

1.1 This specification covers one grade of Boxboard for use in ammunition.

* 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition, General Specifications for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

FSC: 9330

Source: https://assist.dla.mil -- Downloaded: 20261O-19T07:52Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS PUBLICATIONS

D548-71 -	Water Soluble Acidity or Alkalinity of Paper
D641-65 -	Conditioning Paperboard, Fiberboard and Paperboard Containers for Testing
D644-70 -	Moisture Content of Paper and Paperboard by Oven Drying
D645-67 -	Thickness of Paper and Paperboard
D646-67 -	Basis Weight of Paper and Paperboard
D828-71 -	Tensile Breaking Strength of Paper and Paperboard
D2529-68-	Bursting Strength of Paperboard and Linerboard

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103).

* 3. REQUIREMENTS

3.1 Material. The boxboard shall consist of chipboard, newsboard, reclaimed kraft board, or waste paper board, with a 100 percent sulphate kraft liner on one side.

3.2 Color. The boxboard shall be essentially gray to tan, but may have variations in hue and shade and may contain specks, as might be caused by fabricating from mixed waste papers.

3.3 Finish. The boxboard finish shall be calender No. 3 to No. 4 similar to a standard sample furnished or approved by the contracting officer.

3.4 Form and dimension. The boxboard shall be furnished in flat cut sheets. The sheets shall be of the thickness, length and width specified below:

(a) Thickness - Total 0.018 to 0.021 inch (see 4.4.1) Liner, 0.007 to 0.008 inch.

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:52Z
Check the source to verify that this is the current version before use.

(b) Sheet Size- The sheets shall measure 24 inches by 36 inches, the tolerance being plus or minus (±) 1/16 inches, unless otherwise specified in the contract. The longer dimension shall be in the machine direction (see 6.5).

3.5 Basis weight. The minimum weight for 500 sheets, measuring 24 X 36 inches shall be 196 pounds (lbs.) (see 4.4.1).

3.6 Edge. One edge of each sheet in the machine direction shall be deckle-edged (see 6.6).

3.7 Tensile strength. The tensile strength of the boxboard shall meet the following requirements when tested as specified in paragraph 4.4.1:

(a) Machine direction - minimum (min.) 70 lbs.

(b) Cross section - minimum 16 lbs.

3.8 Tensile strength after forming. The loss in tensile strength after forming shall not exceed 5 percent (%) (see 4.4.2).

3.9 Bursting strength. The boxboard shall be capable of meeting the minimum bursting strength of 100 points when tested as specified herein (see 4.4.1).

3.10 Moisture content. The moisture content of the boxboard shall be between the limits of 4 percent and 7 percent, when tested as specified in paragraph 4.4.1.

3.11 Acidity or alkalinity. The boxboard shall have a pH between 6.0 and 7.5, when tested in accordance with paragraph 4.4.1.

3.12 Workmanship. The boxboard shall be free from pieces of unmacerated paper larger than 1/4 square inch in area, wax, asphalt, metallic or abrasive materials, slime, blisters, holes, tears, scuffs, wrinkles and delamination. Trimmed edges shall be clean cut.

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:52Z
Check the source to verify that this is the current version before use.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product. At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Specification number and date, together with an identification and date of changes thereto.

c. c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates, unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation. The term "inspection lot", as used in this specification, is defined as an essentially homogeneous

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:52Z
Check the source to verify that this is the current version before use.

collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same specification and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process or specification not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity criteria specified herein, regardless of the type of sampling procedure which is being applied to determine conformance with requirements.

4.2.2 Lot size. A lot shall consist of all the boxboard of one nominal thickness (not to exceed 20,000 pounds), which has been manufactured from the same types and proportions of ingredients by the same process, by one manufacturer, and submitted for acceptance at one time.

4.2.3 Examination. The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.2) of boxboard. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.10.

4.3 Testing

4.3.1 Sampling. The sampling plans for each of the following tests shall be in accordance with Standard MIL-STD-105, Table IIA, using Code Letter D and an AQL of 0.40 for Major defects and an

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:52Z
Check the source to verify that this is the current version before use.

AQL of 0.65 for each Minor defect:

CLASSIFICATION OF DEFECTS

Category	Defects	Method of Inspection	Code No. (see 6.3)
Color (see 3.2)	Minor	Visual	01001
Finish (see 3.3)	Minor	Visual	01002
Edge (see 3.6)	Minor	Visual	01003
Workmanship (see 3.12)	Minor	Visual	01004

4.3.2 Form and dimension (see 3.4), Minor defect, Code No. 02001. Samples to be tested shall be obtained in accordance with the provisions of ASTM D645-67. Failure of the sample to meet the applicable requirement when tested as specified in 4.4.1 shall be cause for rejection of the lot (see 4.3.9).

4.3.3 Basis weight (see 3.5), Major defect, Code No. 03001. Samples to be tested shall be selected as specified in ASTM D-646-67. Failure of the sample to meet the applicable requirement when tested as specified in 4.4.1 shall reject the lot (see 4.3.9).

4.3.4 Tensile strength (see 3.7), Major defect, Code No. 04001. Test specimens shall be selected in accordance with the provisions of ASTM-D-828-71. Failure to meet the applicable requirement when tested as specified in 4.4.1 shall be cause for rejection of the lot (see 4.3.9).

4.3.5 Tensile strength after forming (see 3.8), Major defect, Code No. 05001. Five samples measuring 12 inches long in the machine direction and 6 inches wide shall be selected for this test. Failure of the sample to meet the applicable requirement shall be cause for rejection (see 4.3.9).

4.3.6 Bursting strength (see 3.9), Major defect, Code No. 06001. Samples shall be selected in accordance with the provisions of ASTM-D2529-68. Failure to meet the applicable requirement when tested as specified in 4.4.1 shall be cause for rejection of the lot (see 4.3.9).

4.3.7 Moisture content (see 3.10), Major defect, Code No. 07001. Samples shall be selected as specified in ASTM-D644-70.

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:52Z
Check the source to verify that this is the current version before use.

Failure of the values of the specimens to meet the applicable requirement shall be cause for rejection (see 4.3.9).

4.3.8 Acidity or alkalinity (see 3.11), Major defect, Code No. 08001. Samples shall be selected as specified in ASTM-D548-71. Failure of the sample to meet the applicable requirement shall be cause for rejection of the lot (see 4.3.9).

4.3.9 Conditioning. Unless otherwise specified, all specimens shall be brought to a substantial equilibrium in a circulating atmosphere maintained at a relative humidity and a temperature specified in ASTM-D 641-65.

4.3.10 Inspection equipment. Inspection equipment required to perform the inspection prescribed in this specification is identified in the Examination (4.2.3) and Test Method (4.4) paragraphs herein. See 6.4 for details concerning responsibilities for inspection equipment design and approval.

4.4 Test methods and procedures

4.4.1 Performance of tests. The following tests shall be performed in accordance with the pertinent document and method specified:

Test	Document	Method
Form and dimensions	ASTM	D645-67
Basis Weight	ASTM	D646-67
Tensile strength	ASTM	D828-71
Bursting strength	ASTM	D2529-68
Moisture content	ASTM	D644-70
Acidity or Alkalinity	ASTM	D548-71

4.4.2 Tensile strength after forming. Wrap a sample of the material, 12 inches long in the machine direction and 6 inches wide around a mandrel 1 1/4 plus or minus (±) 1/8 inch in diameter so that the machine direction is parallel to the axis of the mandrel and the smoother surface of the board is on the outside. Avoid any creasing or folding while winding. Remove the samples by unwinding and test for tensile strength in cross direction. Calculate the percent loss in tensile strength caused by this operation:

$$\text{Percent loss in tensile strength} = \frac{(A-B) \times 100}{A}$$

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:52Z
Check the source to verify that this is the current version before use.

Where:

A = tensile strength before forming (4.3.4)
B = tensile strength cross direction after forming

* 5. PREPARATION FOR DELIVERY

5.1 Packing and Marking

5.1.1 Packing.-Unless otherwise specified, packing shall be Level C.

5.1.1.1 Level C. Boxboard shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to mode of transportation.

5.2 Marking for shipment. Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129. The nomenclature shall be as follows:

BOXBOARD

COLOR_____ WIDTH_____ LENGTH_____
SPECIFICATION MIL-B-20467A (PA)

* 6. NOTES

6.1 Intended use. The boxboard covered by this specification is designed for use in the manufacture of tubes, cases, discs, closing plugs, etc., for Signals, Flares, Electric primers, Cartridges and other components of various end items.

6.2 Ordering Data. Procurement documents shall specify the following:

a. Title, number and date of this specification
b. Size of sheets (if other than 24 inches by 36 inches)
c. Machine direction with relation to sheet dimensions

6.3 Inspection Code Numbers. The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government. These code numbers are

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:52Z
Check the source to verify that this is the currentversion before use.

also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.4.3).

6.4 Inspection equipment designs. Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) and contractor designs. SIE designs are designated by drawing numbers (on the Equipment Lists referenced on the Equipment Tabulation) or (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.4.2 and 6.4.3 should he elect to do so.

6.4.1 SIE designs. SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.4.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.4.1c). Designs shall be of the category

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T07:52Z
Check the source to verify that this is the current version before use.

and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.4.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to: Picatinny Arsenal, ATTN: SARPA-QA-A-P, Dover, New Jersey 07801, in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (Dd Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.5 Machine direction determination. To determine whether the machine direction is the longer dimension, wet one side of a strip about 2 inches square. The strip will curl around an axis parallel to the machine direction.

6.6 Deckle-edge. A deckle-edge is a rough untrimmed edge of the paper formed in a hard mold or by the deckle strap of the machine, and is thinner than the rest of the sheet.

6.7 Marginal Notations. The margin of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

CUSTODIAN:
Army-PA

PREPARING ACTIVITY:
Army-PA

Project Number: 8140-A116

Source: https://assist.dla.mil -- Downloaded: 202610-19T07:52Z
Check the source to verify that this is the current version before use.